SMOOTH CONCRETE PLANTER

Step-by-Step DIY Guide

Cast a Beautiful Round Concrete Pot with a Perfect Smooth Finish

The finished smooth concrete planter with rosemary — in a garden at golden hour

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About This Project

This smooth concrete planter has a finish so clean and refined it looks like it came from a Scandinavian design boutique. Using a two-part plastic mold and fine-grain concrete, you cast a perfectly round, tapered pot with smooth walls — no texture, no rough edges, just clean minimal concrete. Plant it with herbs, succulents, or ornamental grasses for a modern garden accent that costs under $15 to make and sells for $60–100 in design stores.

Difficulty	Easy
Active time	30–40 minutes
Cure time	24–48 hours
Pot size	~20–25 cm diameter, ~22 cm tall (depends on mold)
Cost to make	$5–12 USD
Retail equivalent	$60–120 USD

Materials & Tools

#	Item	Notes
1	Fine-grain concrete or self-compacting concrete	The secret to a smooth finish is fine-grain mix — coarse builders concrete leaves a ro
2	Two-part round plastic mold	An outer mold (defines the outside of the pot) and inner mold (creates the hollow). Pu
3	Water	Clean tap water. Ratio: 2.5 parts concrete to 1 part water for a pourable mix.
4	Mixing bucket + trowel	A 5-litre bucket and mason's trowel for mixing.
5	Cooking spray / mold release	Coat ALL mold surfaces before every pour — essential for the smooth release.
6	Rubber mallet	For tapping the mold to release air bubbles after pouring.
7	Sandpaper (220–400 grit)	For smoothing any minor imperfections after demolding.
8	Concrete sealer (matte)	2 coats — protects the smooth surface and deepens the grey tone.
9	Drill + 12mm bit	For drainage holes if using as a planter.
10	Nitrile gloves	Always wear when handling wet concrete.

■ ***Key to smooth finish:*** *Fine-grain or self-compacting concrete + thorough mold release + slow, careful demolding. These three things make the difference between a smooth design piece and a rough garden pot.*

Step-by-Step Instructions

Step 1 — Mix the Concrete

Mixing fine-grain concrete to a smooth, lump-free consistency

Mix fine-grain concrete with water to a smooth, pourable consistency — like thick cream. The ratio is 2.5 parts dry concrete to 1 part water. Mix thoroughly for 3 minutes until completely lump-free. For the smoothest finish, use self-compacting concrete (SCC) which flows into all areas of the mold without needing vibration.

- Add water gradually — the right consistency pours slowly but doesn't splash.
- Mix in a clean bucket — old concrete residue can cause lumps and surface defects.
- For a coloured pot: add concrete pigment now and mix until uniform throughout.
- Tap the bucket on the table several times to release any trapped air before pouring.

> ■ ***Tip:*** *A slightly wetter mix (more water) flows better into the mold and reduces air bubbles — giving a smoother final surface. Just don't go too wet or the concrete weakens.*

Step 2 — Prepare & Fill the Mold

Pouring concrete into the outer mold — the inner mold sits inside creating the hollow

Coat every surface of both mold parts generously with cooking spray or petroleum jelly. Place the outer mold on a flat surface, then position the inner mold centrally inside it — the gap between the two creates the walls of your planter. Pour the concrete slowly into the gap.

- Pour in a thin, steady stream — avoid splashing which traps air.
- Fill to the top of the mold gap in one continuous pour.
- If the inner mold floats up: weigh it down with a heavy stone or fill it with water.
- Tap the outer mold firmly for 60 seconds after pouring to release all air bubbles.

> ■ ***Critical step:*** *The 60 seconds of tapping is what gives you a bubble-free surface. Any air bubble against the mold wall creates a pit in the finished surface — tap it out while the concrete is still fluid.*

Step 3 — Smooth the Top & Cure

Pressing the mold flat and smooth — concrete fills the entire gap evenly

Once poured and tapped, use a trowel or flat edge to scrape the top of the mold level. This becomes the base of the planter. Cover loosely with plastic and leave completely undisturbed for 24–48 hours.

- Scrape the top flush with the mold rim — excess concrete creates an uneven base.
- Cover with plastic to slow curing and prevent surface cracking.
- Keep in shade — direct sun dries the surface too fast.
- Do NOT move or touch the mold during curing.

> ■ ***Tip:*** *Wait the full 48 hours. The smooth outer surface of the pot needs extra time to develop strength before you peel the mold away.*

Step 4 — Demold — The Reveal!

Lifting the outer mold away — the smooth concrete pot is revealed beneath

After 48 hours, carefully remove the outer mold first — it should peel away cleanly to reveal the smooth exterior. Then remove the inner mold by twisting gently and pulling upward. The moment the mold comes off is one of the most satisfying in all of DIY.

- Remove the outer mold first — flex it gently from all sides before pulling.
- For the inner mold: twist slightly while pulling upward — it should slide out.
- If stuck: wait another 6 hours. Force-demolding causes chipped edges.
- Inspect the surface — any minor pits can be filled with fresh concrete slurry.

> ■ ***What you'll see:*** *A perfectly smooth, clean concrete surface with a subtle grey tone. The smoothness is what makes this pot feel premium — it looks machine-made but you cast it by hand.*

Step 5 — Sand, Drill & Seal

Sand any minor rough spots with 220-grit wet sandpaper. Drill one or two drainage holes in the base with a 12mm concrete bit. Wipe clean and let dry 24 hours before sealing.

- Wet-sand for a cleaner result with less dust — dip the sandpaper in water.
- Drill drainage holes slowly at low speed — let the drill do the work.
- Apply 2 coats of matte concrete sealer — this deepens the grey and protects the surface.
- Let sealer dry completely (24 hours) before planting.

The Finished Planter

The finished smooth concrete planter with rosemary — at golden hour in the garden

Pro Tips & Variations

★ Use self-compacting concrete for perfection	SCC flows into every part of the mold without needing vibration or tapping. The result is virtually bubble-free — a glass-smooth surface straight from the mold. Available from specialist concrete suppliers online.
★ Try white cement for a Nordic look	White Portland cement instead of grey creates a bright, clean pot that looks incredible with dark soil and green plants — a very popular Scandinavian aesthetic.
★ Charcoal for a dramatic look	Add black pigment (2% by weight) for a near-black finish. Charcoal concrete pots with silver or white succulents are one of the most pinned DIY combinations on Pinterest.
★ Make a matching set in 3 sizes	Use the same mold technique with different-sized containers for small, medium and large pots. A set of three in graduated sizes on a shelf or table is a stunning interior display.
★ Polish for a premium finish	After sealing, apply a coat of concrete wax or furniture wax and buff with a soft cloth. This gives the surface a subtle depth and sheen that makes it look genuinely expensive.
★ No mold? Use nested buckets	An outer plastic bucket + smaller inner bucket = improvised mold. Any round or square container combination works. The outside surface won't be as smooth as a purpose-made mold, but it still produces a great pot.
★ Sell at markets	Smooth concrete pots sell for $25–90 at craft markets depending on size. A matched pair retails for $60–140. Cost to make: $5–15 each. Excellent margin with repeat buyers.
★ Add a lip or rim detail	Before the concrete fully cures (~4 hours), press a strip of cardboard or foam around the top edge to create a raised rim detail. Remove after curing — the clean line looks incredibly refined.

Troubleshooting

✗ Problem: Surface has pits and bubbles

✓ **Solution:** Didn't tap the mold enough after pouring. Next time: tap firmly for a full 60 seconds on all sides. Also try a slightly wetter mix that flows better. Fill existing pits with cement slurry and sand smooth once dry.

✗ Problem: Pot stuck in the mold

✓ **Solution:** Didn't use enough mold release, or demolded too early. Wait 48 hours. Flex the mold vigorously — push it inward from all sides. As a last resort, cut the outer mold away.

✗ Problem: Vertical crack appeared

✓ **Solution:** Concrete dried too fast or the inner mold was removed too violently. Keep moist during curing. Remove inner mold slowly and gently. Hairline cracks can be filled with cement slurry.

✗ Problem: Surface looks dull/chalky after sealing

✓ **Solution:** Applied sealer to damp concrete. Always let concrete dry completely (24h minimum) before sealing. Sand lightly and apply a fresh coat of sealer to dry concrete.

✗ Problem: Pot feels fragile at the rim

✓ **Solution:** Walls are too thin at the top. Ensure minimum 15mm wall thickness. Thicken by adjusting the inner mold position, or reinforce with a fresh coat of cement slurry on the inside.

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